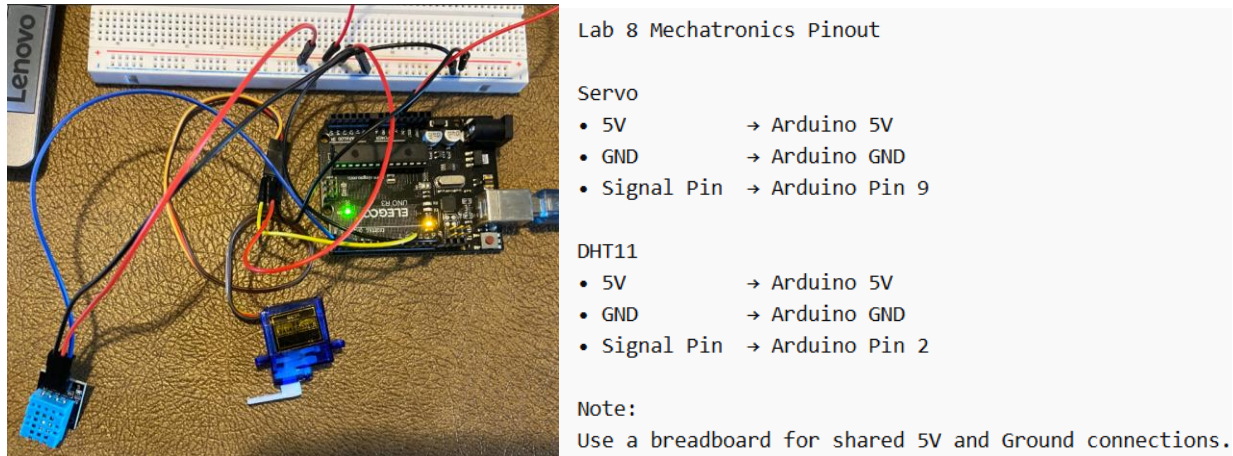


Lab 8 Option 2 Report DIY Mechatronics System

Section 1

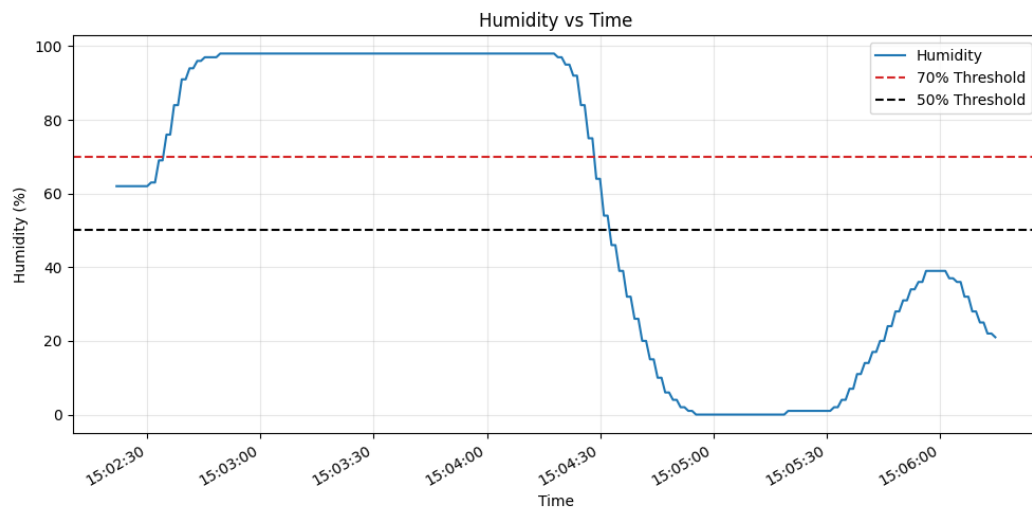
Below is the image of my DIY mechatronics system with a pinout description:



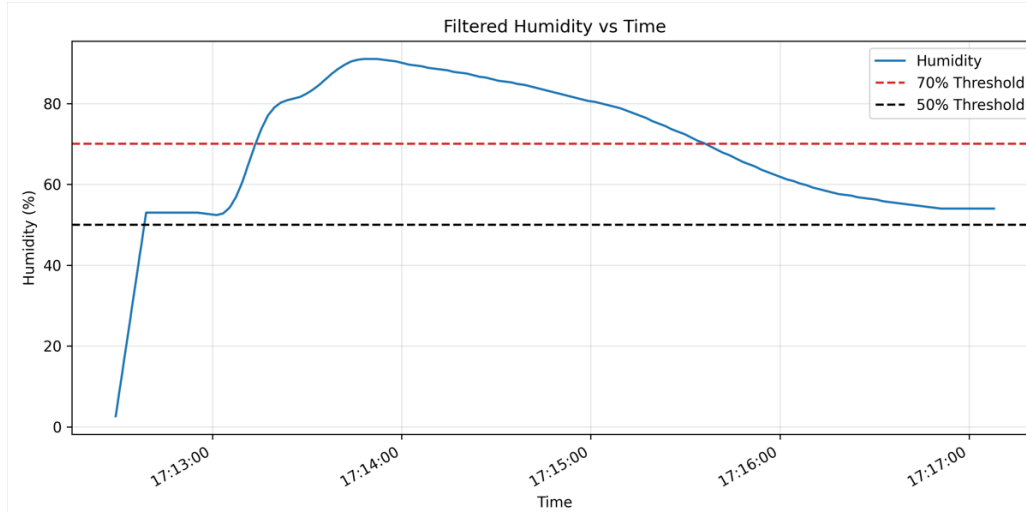
This mechatronics system utilizes a servo, a DHT11 sensor, and an Arduino. The system reads humidity levels using the DHT11 and will adjust the servo angle according to the humidity. In practice, this system is to be used in a greenhouse, which needs to maintain a humidity of around 60%. The servo is to be attached to a vent, which will open and close (controlled by servo angle) according to the humidity of the greenhouse to maintain a humidity of 60%. I chose to make this system to apply the “niche” humidity sensor in the real world. I felt that an automated system would be helpful for gardening, and keeping plants healthy in a greenhouse without the need of supervision.

Section 2

Below is the raw, unfiltered data of the DHT11 sensor:



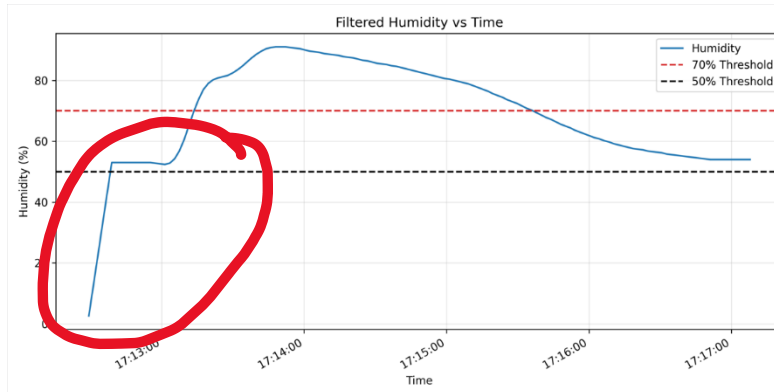
Below is the filtered data of the DHT11 sensor, filtered with a Simple Moving Average filter.



I chose a simple moving average filter for this data because it is a simple filter. The humidity of a greenhouse will likely not change at an instant, so multiple data points may be used as reference to smooth out the data. I believe it to be working well as I can see that the data in the filtered plot is much smoother. The filtered data plot leaves one issue to be addressed which I will specify in section 3.

Section 3:

Below is a correction in my system code that I had to make. In the filter function “SMA<...>”, I had previously set “average = {0.0}”. This was adjusted to be “60.0”.



```
Complete_code_option2.ino
1 #include <Adafruit_Sensor.h>
2 #include <DHT.h>
3 #include <DHT_U.h>
4 #include <Filters.h>
5 #include <Filters/SMA.hpp>
6 #include <Servo.h>
7
8 #define DHTPIN 2
9 #define DHTTYPE DHT11
10 #define SERVO_PIN 9
11
12 DHT_Unified dht(DHTPIN, DHTTYPE);
13 Servo windowServo;
14
15 uint32_t delayMS;
16 SMA<S, float, float> average = {60.0};
17
18 void setup() {
19   Serial.begin(9600);
20   dht.begin();
21   windowServo.attach(SERVO_PIN);
22   delayMS = 2000;
23 }
24
```

This correction was made in order to adjust the steep incline in the data at the beginning of recording. This steep incline forces the servo to adjust position, though the steep incline in data does not exemplify the real humidity. The steep incline happens when the SMA function average is equal to zero, because this means that an average between the raw data, and the setpoint of zero is being recorded for the initial data reads. Adjusting this value to “60.0” ensures that the data and therefore the servo starts at a reasonable position.

Section 4

- I used no online sources
- I used no in class collaborators

Section 5

- I expected in this lab to learn how to create a mechatronics system in the real world. I expected to create a system which I could use and build myself in the future.
- I would rate my attendance as excellent in this lab. I attended lab enough to get all check offs for both lab options (extra credit). No adjustments needed.
- I did well with this lab assignment. I was able to combine aspects of multiple labs to create a system from scratch which can operate in the real world and be a useful tool for the gardening community. I give myself a 4/4 on this lab.
- I learned how to apply much of the lab section knowledge into a project. I learned how to plan out a project, find customer need, and develop a solution.

